WannaCry Ransomware Attack — Intelligence & Detection Report

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EXECUTIVE SUMMARY

On 12 May 2017 at 07:44 UTC, a ransomware cryptoworm designated WannaCry began propagating across global networks, infecting 230,000+ systems across 150+ countries within approximately seven hours. Estimated global damages exceed $4 billion USD. Total ransom collected was approximately $130,000 — a ratio that raises significant questions about the operation's true intent.

WannaCry exploited CVE-2017-0144, a critical heap buffer overflow vulnerability in Microsoft's SMBv1 protocol rated 9.3 (CVSS v2). Microsoft had released patch MS17-010 on 14 March 2017 — 59 days before the attack. Organizations that applied the patch were fully immune. Most did not.

The attack required no user interaction. From a single internet-exposed port 445, the full kill chain — exploitation, payload injection, persistence establishment, encryption, and propagation — executed automatically within minutes on each new host.

Attribution: Lazarus Group (assessed moderate-to-high confidence) with suspected DPRK nexus (assessed moderate confidence), based on multi-vendor consensus from Microsoft, NSA, GCHQ, and the 2018 US Department of Justice indictment of Park Jin Hyok. The disproportionate damage-to-ransom ratio and the presence of a hardcoded kill-switch domain — assessed as a QA artifact rather than an intentional design — suggests the malware was deployed before quality assurance was complete, indicating either a poorly executed financial operation or a dual-purpose disruption campaign with ransomware framing.

ATTACK CHAIN SUMMARY

The WannaCry attack executed across six sequential stages:

Stage 1 — Initial Exploitation: EternalBlue (leaked NSA tooling) exploited CVE-2017-0144 over TCP port 445 to achieve unauthenticated kernel-level remote code execution on unpatched Windows systems. No user interaction required.

Stage 2 — Payload Injection: DoublePulsar, a kernel-mode backdoor also leaked by The Shadow Brokers, was injected into memory to deliver the ransomware loader while avoiding disk-based antivirus detection.

Stage 3 — Loader Execution & Persistence: The ransomware binary tasksche.exe was written to C:\Windows\ and executed. A Windows service named mssecsvc2.0 — mimicking the legitimate "Microsoft Security Center" — was created for persistence across reboots. Files were hidden using attrib +h and permissions modified via icacls to resist removal.

Stage 4 — Kill-Switch Check: The ransomware module tasksche.exe performed an HTTP GET request to a hardcoded domain before initiating encryption. If the domain resolved, execution halted. If unreachable, the attack proceeded. This check is confirmed to exist within the ransomware module only — the worm propagation component continued independently regardless of kill-switch status.

Stage 5 — Encryption: Shadow copies were deleted using vssadmin delete shadows /all /quiet, bcdedit /set {default} recoveryenabled no, and wbadmin delete catalog -quiet before encryption began — eliminating all native recovery options. Over 170 file types were encrypted using AES-128-CBC per file, with each AES key protected by RSA-2048. Encrypted files received .WNCRY, .WCRY, or .WNCRYT extensions. A ransom demand of $300–$600 in Bitcoin was displayed via @WanaDecryptor@.exe.

Stage 6 — Worm Propagation: Two simultaneous scanning threads — one targeting local LAN subnets, one targeting random internet IPs — ran in parallel with encryption. Each newly infected host became an independent propagation node. This architecture enabled thousands of exploitation attempts per minute, accounting for the 230,000-system infection rate within a single day.

KEY DETECTION SIGNALS

Signal	Log Source	Event ID	Confidence
mssecsvc2.0 service creation	Windows Event Log	7045	High
tasksche.exe execution from C:\Windows\	Windows Event Log	4688	High
vssadmin delete shadows execution	Sysmon	1	High
Mass .WNCRY file creation	Sysmon	11	High
Kill-switch domain DNS query	DNS logs	—	Confirmed infection
High-volume TCP 445 outbound	Firewall/IDS	—	High
East-west workstation SMB traffic	Network telemetry	—	Medium-High

High-confidence detection: Correlation of any three signals above within a 60-minute window constitutes a confirmed WannaCry incident requiring immediate endpoint isolation.

CRITICAL INTELLIGENCE JUDGMENTS

On the kill switch: The hardcoded kill-switch domain is assessed with high confidence as a development artifact or sandbox evasion mechanism — not an intentional off switch. Its registration by researcher Marcus Hutchins on 12 May 2017 slowed external encryption propagation but did not stop internal network infections or the worm scanning component.

On attribution: The $130,000 ransom collected versus $4–5 billion in damage caused represents an operationally anomalous ratio for a financially motivated campaign. This report assesses with moderate confidence that WannaCry was either a poorly executed financial operation or a dual-purpose campaign using ransomware as cover for disruption objectives aligned with DPRK strategic interests.

On future relevance: EternalBlue-based exploitation was subsequently reused in NotPetya (June 2017) and BadRabbit (October 2017). Legacy systems in healthcare, manufacturing, and critical infrastructure sectors globally

remain exposed to SMB-based exploitation. WannaCry's core attack chain remains reproducible against unpatched environments today.

DEFENSIVE PRIORITIES

Critical (implement immediately): Apply MS17-010 on all Windows systems. Disable SMBv1 where not operationally required. Block TCP 445 and TCP 139 at the network perimeter.

High (implement within 30 days): Deploy network segmentation preventing workstation-to-workstation SMB communication. Establish offline immutable backups following the 3-2-1 rule. Enable Sysmon with process creation and file event logging.

Ongoing: Correlate endpoint alerts with network telemetry in SIEM. Maintain asset inventory with continuous vulnerability scanning. Monitor threat intelligence feeds for EternalBlue variant exploitation.

REPORT SCOPE

This report covers: attack timeline, six-stage technical execution chain, CVE-2017-0144 vulnerability analysis, Microsoft Sentinel KQL and Splunk detection queries, YARA detection rule, full IOC table (hashes, domains, registry keys, mutex, Bitcoin wallets), complete MITRE ATT&CK mapping with 12 confirmed techniques, incident response procedures based on NIST SP 800-61, layered prevention architecture, and intelligence assessment with attribution analysis.

Analyst Confidence: High for technical analysis. Moderate for attribution. Low for intent assessment. Attribution relies on published analysis from Microsoft, NSA, GCHQ, and US DoJ. No independent attribution analysis was performed.